COS 201 FINAL ASSIGNMENT

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QUESTION 1

Create a typical array and demonstrate how to reference the elements in it using any preferred examples.

ANSWER 1 — Demonstration of Java Arrays

In Java, an array is a data structure used to store multiple values of the same type in a single variable. Arrays have a fixed size and can be accessed using an index that starts from 0.

Example: Declaring and Using an Array

```
public class ArrayExample {
    public static void main(String[] args) {

        // Declaring and initializing an array of integers
        int[] scores = {85, 90, 76, 88, 95};

        // Accessing elements using their index
        System.out.println("First score: " + scores[0]);
        System.out.println("Third score: " + scores[2]);

        // Modifying an element
        scores[1] = 92;

        System.out.println("Updated second score: " + scores[1]);

        // Looping through the array
        System.out.println("\nAll scores:");
        for (int i = 0; i < scores.length; i++) {
```

```
            System.out.println("Index " + i + ": " + scores[i]);
        }
    }
}
```

QUESTION 2

Demonstrate how to use collection class ArrayList<T>, section 10.9, and implement using your own chosen example.

ANSWER 2 — Demonstration of ArrayList<T>

The ArrayList<T> class belongs to Java's Collections Framework. It grows dynamically and provides built- in methods such as add(), get(), set(), size(), and remove().

Example: Using ArrayList in Java

```
import java.util.ArrayList;

public class ArrayListExample {
    public static void main(String[] args) {

        // Creating an ArrayList of Strings to store student names
        ArrayList<String> students = new ArrayList<>();

        // Adding elements to the ArrayList
        students.add("Charles");
        students.add("Daniel");
        students.add("Chinedum");

        // Displaying the ArrayList
        System.out.println("Student List: " + students);

        // Accessing elements
        System.out.println("First student: " + students.get(0));

        // Modifying an element
        students.set(1, "David");
```

```
        System.out.println("Updated List: " + students);

        // Removing an element
        students.remove(2);

        System.out.println("List after removal: " + students);

        // Looping through the ArrayList
        System.out.println("\nAll Students:");
        for (String name : students) {
            System.out.println(name);
        }
    }
}
```